
Mind2Dialogue: Training Human-Aware Language Models through Shared-State User Simulation

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Project



Code



Dataset

Abstract

Personalized assistants must infer users’ unobserved beliefs, goals, affect, and social relationships from interaction histories, yet training data rarely provide direct supervision for these latent states or their evolution. We introduce **Mind2Dialogue**, a scalable simulation framework that generates user–assistant conversations conditioned on evolving structured user states. These states shape both sides of the dialogue: they condition each user message and are observed by an *Oracle* assistant when generating responses. We then train a student model to reproduce Oracle responses from static persona information and dialogue history, enabling it to approximate state-conditioned behavior without access to the evolving state at inference time. Without requiring per-dialogue human annotation, the pipeline produces M2D-CORPUS, which contains synthetic multi-turn dialogues and QA pairs in which assistant responses and answers are grounded in the corresponding latent user states. Fine-tuning Qwen2.5-7B-Instruct on M2D-CORPUS improves PrefEval-Gen by 33.4 absolute points and PersonaMem-v2 by 10.0 points over the corresponding base model; the resulting model outperforms memory-augmented and personalization baselines, and the personalization gains hold across three model families. Although trained without explicit Theory-of-Mind supervision, the Qwen and Llama models also improve on ToM benchmarks. Together, these results suggest that personalization and Theory of Mind may rely on a shared capability: inferring latent user states from observed interactions.¹

1 Introduction

Beyond general-purpose capability, effective assistants must infer and respond to the beliefs, preferences, goals, and needs of the users they serve [37, 13]. For real-world deployment, personalized assistance requires not only solving a task but also doing so in a way that reflects what a particular user wants, believes, and needs. In this paper, we study how this capacity can be developed in model weights through training.

What does it mean, computationally, for a model to understand a person? In this work, we operationalize it as inferring aspects of an unobserved user state from observable interactions: the beliefs, intentions, emotions, and relational context that make a sequence of messages coherent [34, 1]. AI research has approached this problem from two related directions. Personalization focuses on

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¹Code and data are available at <https://github.com/wannabeyourfriend/mind2dialogue>.

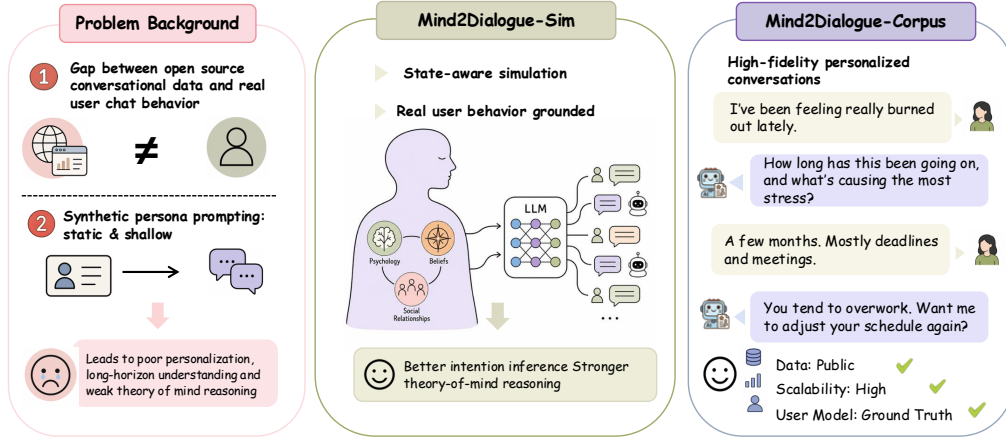


Figure 1: Overview of **Mind2Dialogue**.

understanding *this particular user* [37, 14, 52], while Theory of Mind studies reasoning about the mental states of *any agent* more generally [38, 17, 8, 39]. Although the two settings are distinct, both require reasoning about information that is not explicit in observed messages. We therefore hypothesize that training models to infer evolving user states for personalization may also transfer to Theory-of-Mind tasks and test this hypothesis without using ToM-specific supervision. Because ToM benchmark performance can be sensitive to surface perturbations [45, 41], we interpret any observed transfer narrowly.

Conversations that draw on substantial shared context often occur between people who know one another well, such as close friends, family members, and long-term collaborators. These interactions are difficult to collect at scale because they are typically private and undocumented. Available public data provide only partial substitutes: behavioral logs encode preference signals but offer limited linguistic supervision [25], whereas public dialogue corpora contain natural language but rarely capture relationships and user states accumulated across interactions. Synthetic data offer one way to bridge this gap. Existing synthetic dialogue pipelines, however, typically either condition generation on static profiles or generate assistant responses without direct access to the evolving state that governs the simulated user. Consequently, these data do not directly demonstrate how an assistant would respond if the underlying user state were known.

This creates a supervision gap: training user-state inference would benefit from assistant responses grounded in the user’s state, yet real conversations rarely expose that state. We use the term *Oracle assistant* to denote a teacher that directly observes the simulator-defined user state when producing a response. We therefore address two questions: (i) how can an Oracle assistant be instantiated within a scalable data-generation process, and (ii) how can its state-conditioned behavior be transferred to a deployable model that observes dialogue history but not the persona or evolving state used during simulation?

We propose **Mind2Dialogue**, a framework that constructs this information asymmetry in simulation. Its central design is to expose the same evolving structured user state to both the user simulator and the Oracle assistant; the state is privileged information for the student but is not hidden from the Oracle. To address (i), we build M2D-SIM, in which beliefs, intentions, affect, and relational context are updated over the course of an interaction. The state conditions each user utterance and is provided to the Oracle when it generates the corresponding assistant response, yielding state-grounded training targets. To address (ii), we fine-tune a student to reproduce Oracle responses from dialogue history alone. The objective is standard supervised fine-tuning; the asymmetry lies in the supervision rather than in the algorithm, since each target was produced by an assistant that could read a state the student never observes. The student is trained to approximate state-conditioned behavior without access to the structured state at inference time.

We evaluate **Mind2Dialogue** end-to-end on five benchmarks. PersonaMem-v1 [13], PersonaMem-v2 [14], and PrefEval [52] measure personalization, while BigToM [8] and ToMi [18] serve as transfer tests of benchmark-defined Theory-of-Mind reasoning. M2D-CHAT (Qwen2.5-7B-Instruct fine-

tuned on M2D-CORPUS) improves PrefEval generation by 33.4 points and PersonaMem-v2 by 10.0 points over the base model, surpassing both Mem0 and prior personalization-data baselines under the same backbone. The same checkpoint improves BigToM Forward Belief by 13.0 points and Forward Action by 7.5 points, exceeding AutoToM and Thought-Tracing even though M2D-CORPUS was not constructed with explicit Theory-of-Mind supervision, false-belief tasks, or third-person belief annotations. Personalization gains replicate across Qwen, Llama, and OLMo, whereas the Theory-of-Mind transfer reproduces on Qwen and Llama, indicating that this transfer is backbone-dependent. A horizon study compares the full simulator with an ablation that removes both state maintenance and behavior prompting: the resulting dialogues are initially similar at turn 0 but progressively diverge in judged persona consistency by turn 40. Qualitative examples show that the Oracle responds to states not explicitly articulated by the user, including unspoken concerns, identity-relevant considerations, and implicit goals, while adopting the user’s evaluative frame and vocabulary. The student does not observe the evolving states directly; it is trained only on dialogue histories, which may contain manifestations of those states, paired with the Oracle responses. Taken together, the results establish robust personalization gains and provide evidence that state-grounded personalization supervision can transfer to some belief-inference tasks. This transfer is consistent with a partially shared latent-state inference capability, but does not by itself establish a general or human-like Theory of Mind.

2 Related Work

User simulation for personalization Personalized language models either augment a fixed model with external memory and retrieval [3, 54, 19] or adapt model parameters using user-specific data [37, 27]. Synthetic dialogue research ranges from fixed persona-conditioned generation [50, 9] to multi-turn and multi-session LLM user simulators [4, 40, 29, 14]. Some recent simulators also model changing user states [53, 6, 24]. Our distinction is therefore not statefulness alone, but the information structure used to generate supervision: in **Mind2Dialogue**, the user simulator and Oracle share the simulator-defined evolving state, while the student learns from the Oracle responses without observing that state.

Theory of Mind and privileged information. Theory-of-Mind benchmarks evaluate belief and mental-state tracking, but their results can be sensitive to surface perturbations [45, 41]. We therefore use them as transfer tests. Methodologically, our teacher–student asymmetry instantiates learning with privileged information and generalized distillation [46, 11, 21]; the transfer itself uses standard supervised fine-tuning rather than a new distillation algorithm. Section C provides detailed comparisons with these research areas.

3 Mind2Dialogue: A Psychology-Inspired Framework for Stateful User Simulation

Mind2Dialogue generates multi-turn training data with a user simulator and an Oracle assistant. During data generation, both components have access to a persona p and a shared structured state s_t . At each turn, a state-update policy first produces the next state, a user policy produces the next user message, and an Oracle policy produces the target response from the updated dialogue history. The student is trained to predict the Oracle response from the dialogue history alone; neither p nor s_t is included in the student input. This design separates information used to generate the data from information available to the student.

3.1 Problem Formulation

Let p denote the persona used to generate a dialogue, $H_{<t}$ the dialogue history before the t -th user message, and m_t the user message at that turn. We define

$$H_t = (H_{<t}, m_t)$$

as the dialogue history through the user message. The student observes only H_t when predicting the corresponding assistant response.

At the start of turn t , the user simulator produces the next structured state:

$$s_t \sim \pi_{\text{user}}(\cdot \mid p, s_{t-1}, H_{<t}).$$

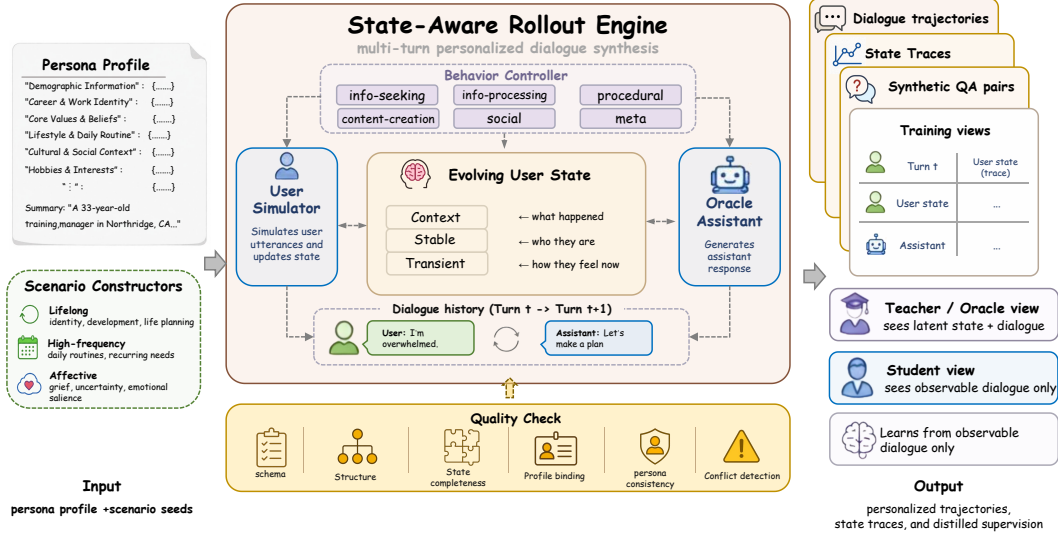


Figure 2: The **Mind2Dialogue** simulation framework. A stateful user simulator and an Oracle assistant are coupled through an evolving latent state s_t ; the student is trained without access to s_t .

Then the user generates a message from the updated state:

$$m_t \sim \pi_{\text{user}}(\cdot \mid p, s_t, H_{<t}).$$

After the message is added to the dialogue history, the Oracle generates the target response:

$$a_t \sim \pi_{\text{oracle}}(\cdot \mid p, s_t, H_t).$$

From N generated dialogues, we construct the dialogue training set

$$\text{M2D-CORPUS}_{\text{dialogue}} = \left\{ \left(H_t^{(i)}, a_t^{(i)} \right) \mid 1 \leq i \leq N, 1 \leq t \leq T_i \right\},$$

where T_i is the number of assistant-response targets in dialogue i . The data pipeline may retain p and s_t as generation metadata, but neither variable is included in the student-facing input.

3.2 Controlled Dialogue Simulation with Scenarios, State, and Behavior Modes

M2D-SIM controls three parts of user simulation explicitly: the scenario that initiates the dialogue, the structured state that changes across turns, and the behavior mode expressed at each turn. This decomposition is motivated by the person–situation view in personality psychology, which treats observed behavior as a product of both individual and situational factors [7]. The cognitive-affective processing account further describes how situational features interact with goals, affect, expectations, and related internal variables to produce context-dependent behavior [28].

In our implementation, the scenario supplies the immediate context for the interaction, s_t records simulator-defined variables that carry across turns, and dynamic behavior-mode prompting controls how the user acts at each turn. If these elements are left implicit in a single prompt, the LLM must determine all three during generation. M2D-SIM instead exposes them as separate controls. The design targets two observable properties of the generated data: coherence across turns and coverage of different user behaviors. M2D-SIM implements this design shown in Algorithm 1.

Scenario construction. Each dialogue begins with a scenario derived from the persona p . The scenario specifies why the user starts the interaction and provides a setting in which persona-specific information can affect how the assistant should respond. We construct scenarios in three categories: **LIFELONG**, covering identity and long-term personal development; **HIGH-FREQUENCY**, covering recurring everyday needs; and **AFFECTIVE**, covering emotionally significant situations such as grief or uncertainty. We filter candidate scenarios using three criteria: level of abstraction, embedding-based distinctness from existing scenarios, and semantic consistency with the persona. For the distinctness check, we reject a candidate if its cosine similarity to any existing scenario exceeds θ_{sim} . Accepted scenarios are cached for each persona so that subsequent runs can reuse the same scenario set.

Stateful user simulation. M2D-SIM represents the state shared by the generation policies as a structured record:

$$s_t = (c_t, z_t^{\text{stable}}, z_t^{\text{transient}}),$$

where c_t contains dialogue-level context, including the turn index, unresolved goals, trust history, and a summary of the interaction. The component z_t^{stable} stores information that changes slowly, such as user values, background constraints, and the user’s stance toward the assistant. The component $z_t^{\text{transient}}$ stores short-lived information, such as mood, current concerns, and judgments made during the turn. The state contains a summary of prior interactions rather than a copy of the full dialogue history at every turn. These fields are simulator-defined control variables, not measurements of a real user’s mental state.

Dynamic behavior-mode prompting. Without additional guidance, an LLM-based user simulator may produce passive or repetitive interactions. To vary user behavior across a dialogue, we use a behavior controller based on the Taxonomy of User Needs and Actions (TUNA) proposed by Shelby et al. [42]. At each turn, the controller selects one of 16 behavioral modes from six families: information seeking, information processing, procedural guidance, content creation, social interaction, and meta-conversation. It then adds the instructions associated with the selected mode to the user-simulator prompt. The controller selects and injects this guidance separately at each turn rather than fixing one behavior mode for the full dialogue. It provides less behavioral guidance at turn 1 and more guidance at later turns or on turns with greater delegation to the assistant. Its selection procedure also encourages coverage across the six families. The full taxonomy and mode descriptions appear in Section H.

QA-format generation. In addition to dialogue-response targets, we use saved dialogue trajectories to construct question–answer training examples. The item generator receives the persona, the structured state, and an excerpt from the source dialogue, and produces a question–answer pair based on the simulated interaction. These inputs guide example generation but are not included directly in the input provided to the student. We generate multiple-choice persona-memory examples and free-form preference-following examples using the same message schema as the dialogue examples. We ensure that evaluation datasets remain held out by n-gram contamination auditing.

3.3 Student Model Fine-Tuning

We fine-tune a student policy π_ϕ on M2D-CORPUS. Each training example consists of a student-visible input sequence x and a target assistant response y . For dialogue-response examples, $x = H_t$ and $y = a_t$. We minimize the standard autoregressive cross-entropy loss over the target response tokens:

$$\mathcal{L}_{\text{SFT}}(\phi) = -\mathbb{E}_{(x,y) \sim \text{M2D-CORPUS}} \left[\sum_{j=1}^{|y|} \log \pi_\phi(y_j \mid x, y_{<j}) \right]. \quad (1)$$

Only tokens in the target response y contribute to the loss. Oracle has access to p and s_t when producing dialogue targets, so both variables can affect a_t . However, the training objective only teaches the student to predict the target response from the visible input. It neither requires nor verifies that the student reconstructs p or s_t internally. The backbone-dependent Theory-of-Mind transfer reported in Section 4.4 is compatible with the student learning to use relevant evidence in the visible dialogue history, but the behavioral results do not identify the student’s internal representation or inference process.

4 Training Student Models on M2D-CORPUS

4.1 Corpus Coverage

Running M2D-SIM over the persona pool yields M2D-CORPUS seeded from 289 personas. The corpus includes a subset of 6,330 multi-turn conversations. We characterize the corpus along three axes: scenario category, persona, and behavioral mode. Scenario categories and behavioral modes are assigned during generation.

Scenarios. Every conversation carries a scenario category assigned by its constructor. The subset spans 47 categories across the three families introduced above: affective (1,428 conversations),

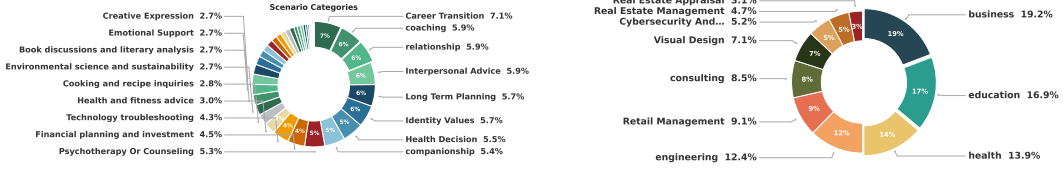


Figure 3: (a) Scenario category distribution. (b) Persona domain expertise distribution.

high-frequency (2,020), and lifelong (2,882). Coverage is long-tailed rather than concentrated on a few topics: the top 9, 18, and 28 categories account for $\sim 52\%$, 81% , and 96% of the data (Figure 3a).

Personas. Each persona declares a free-form `specialization`; the pool contains 153 distinct values, which we cluster into 10 groups for visualization (Figure 3b). Personas are grounded in five focal countries, and the broader rollout corpus extends to a global pool.

Behavior. Each user turn is assigned one of the 16 TUNA modes described above, with coverage enforced across the six families. Figure 4 gives the totals per mode, decomposed by conversation family.

The generation pipeline can be expanded with additional personas, scenario categories, or behavioral modes without per-dialogue human annotation. Section F reports more detailed corpus statistics.

Quality control. We apply four programmatic checks to generated conversations: schema validity, structural sanity (turn count, role alternation, and token bounds), state-trajectory completeness, and profile binding. Two LLM-based checks additionally score persona consistency and flag conflicts with fixed persona attributes, using a judge model distinct from the simulator and the Oracle. Section F.6 describes the checks and reports a human audit of the filtered corpus; the two judge rubrics are represented in Section G.5.

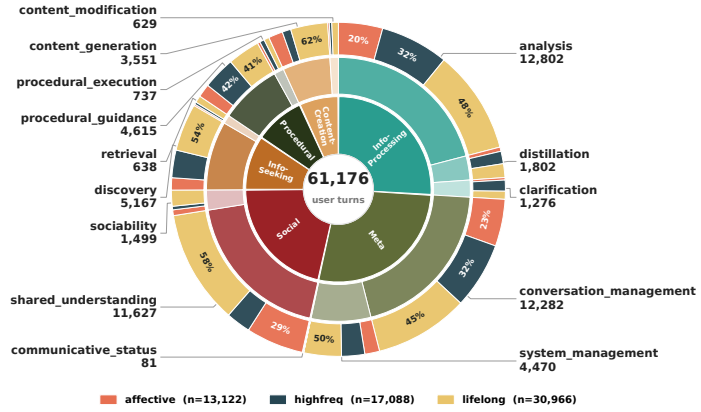


Figure 4: Behavioral mode distribution.

4.2 Pilot Study: M2D-SIM versus Vanilla Simulation

Before full-scale student training, we compare conversations generated by M2D-SIM with those by a **Vanilla** simulator. Both simulators are driven by the same model backbone and are prompted by the same persona in each conversation. Vanilla simulator omits the structured state and the behavior-mode controller. The main comparison therefore measures the combined difference introduced by these two components. Section D.1 reports the per-dimension judge scores and an ablation of a component that evaluates state maintenance and behavior prompting separately.

At turn 0, we do not detect a statistically significant difference between the two conditions (A, D). Across the measured turns, the correlation between turn index and Cohen’s d is $r=0.90$ in the 20-turn study (C) and $r=0.79$ in the 40-turn study (F). For topic depth, the effect size increases from $d=0.16$ at turn 5 to $d=1.22$ at turn 30 (E). The assistant-personalization scores show a similar pattern in the 40-turn study: the Vanilla score decreases from 2.75 to 2.61, while the M2D-SIM score remains near 3.1 (D). Under the judge-based metrics used in this pilot, the difference favoring M2D-SIM is larger at later turns.

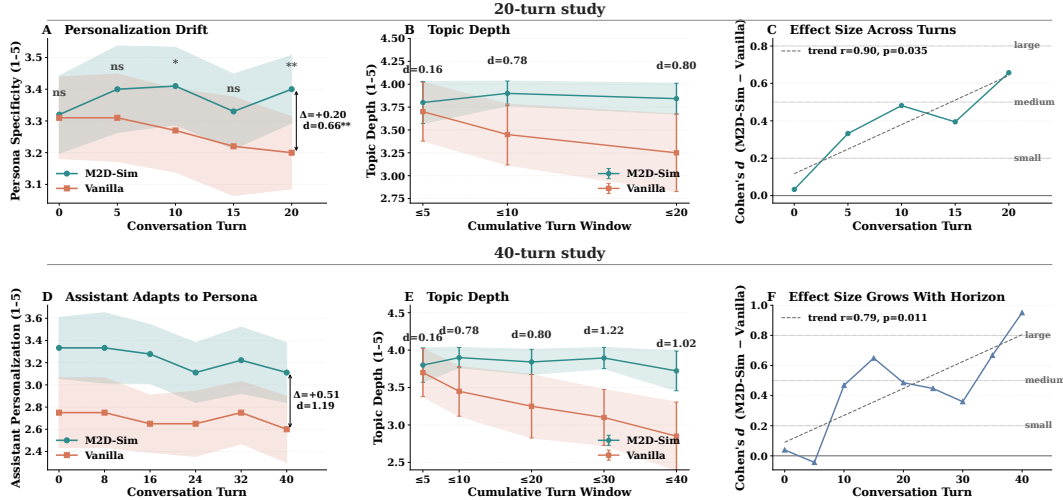


Figure 5: Corpus-level pilot comparison of M2D-SIM and vanilla simulator across dialogue turns in the 20-turn and 40-turn settings.

4.3 Experimental Setup

Benchmarks. We evaluate two benchmark families. The personalization suite contains **PersonaMem-v1** and **PersonaMem-v2** [13, 14], which test long-term persona memory with multiple-choice questions, and **PrefEval** [52], which tests preference adherence through classification and free-form generation. The Theory-of-Mind (ToM) suite contains **ToMi** [18], a short-story belief-reasoning benchmark, and **BigToM** [8], for which we report Forward Belief and Forward Action prediction. The personalization suite measures performance on the target use case. The ToM suite tests whether the learned behavior transfers to tasks outside the supervision used to construct M2D-CORPUS.

Backbone model and baseline methods. Our primary backbone is Qwen2.5-7B-Instruct [35]. We compare M2D-CHAT with the unmodified backbone and with task-specific methods implemented on the same backbone. The personalization baselines are PersonaVLM [30], HumanLM [48], LLMoPt [26], and Qwen2.5-7B-Instruct augmented with Mem0 [3]; the ToM baselines are AutoToM [51] and Thought-Tracing [16]. We include GPT-4o-mini [32] and GPT-5-mini [44] as proprietary reference points. To test whether data scaling depends on the backbone, we also train Llama-3.1-8B-Instruct [10] and OLMo-3-7B-Instruct [31].

Training. We fine-tune each open-source backbone on four fractions of the SFT mixture: $1/8$, $1/4$, $1/2$, and ALL. We keep the training recipe fixed across benchmark families and data scales. Training uses LoRA [12] with 4-bit quantization [5], response-only masking, and AdamW [22] with a cosine learning-rate schedule. Unless noted otherwise, reported gains are absolute changes from the corresponding unmodified backbone.

4.4 Main Results

Performance continues to improve by scaling training data. Larger fractions of M2D-CORPUS produce consistent endpoint gains across model families (Figure 6 and Table 3). With the full mixture, Qwen2.5-7B-Instruct, Llama-3.1-8B-Instruct, and OLMo-3-7B-Instruct improve by +10.0, +13.6, and +10.2 points on PersonaMem-v2 MCQ, and by +33.4, +40.9, and +26.6 points on PrefEval-Gen. PersonaMem-v1 gains are smaller, from +5.6 to +6.5, but remain positive for every backbone. Across all three backbones, the full-data gains are larger for free-form preference generation than for multiple-choice persona recall. On PrefEval-Gen, Qwen2.5-7B-Instruct rises from 23.4 at Base to 36.5 at $1/8$, changes little through $1/2$ (37.9), and then reaches 56.8 with ALL. Llama-3.1-8B-Instruct improves earlier, moving from 24.5 at Base to 56.0 by $1/4$ and 65.4 with ALL. OLMo-3-7B-Instruct improves more steadily, from 52.5 to 79.1. PersonaMem-v1 has smaller and less uniform increments: OLMo-3-7B-Instruct remains at 41.8 from $1/4$ to $1/2$, and Llama-3.1-8B-Instruct moves only from 45.7 to 46.2 over the same interval, although both improve again with the full mixture.

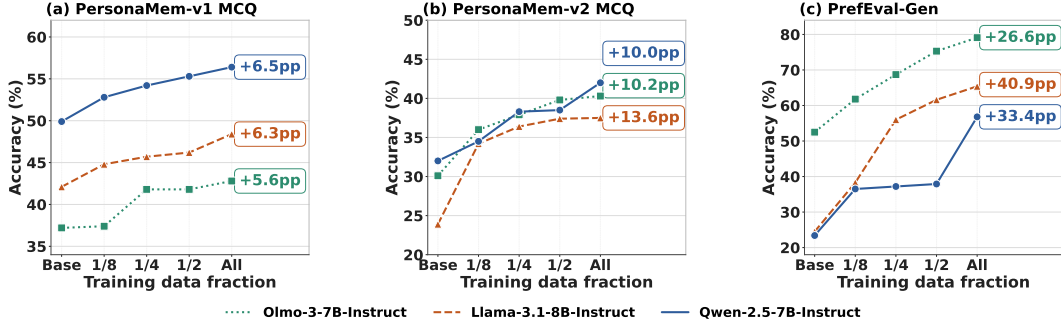


Figure 6: Personalization performance as a function of M2D-CORPUS training data for three open-source backbones. Accuracy is reported on (a) PersonaMem-v1 MCQ, (b) PersonaMem-v2 MCQ, and (c) PrefEval-Gen. Labels report the absolute gain from the unmodified backbone to the full SFT mixture; Table 3 gives the underlying values.

Table 1: Accuracy (%) on personalization benchmarks. **Bold** marks the best non-proprietary result in each column.

Model / Method	PM-v1 MCQ	PM-v2 MCQ	PrefEval Gen	PrefEval Cls
GPT-4o-mini	48.6	37.3	21.1	84.6
GPT-5-mini	61.5	49.2	90.8	99.3
Qwen2.5-7B-Instruct	49.9	32.0	23.4	62.0
PersonaVLM	33.9	24.3	29.8	54.7
HumanLM	38.1	27.3	43.6	78.5
LLMoPt	35.3	22.2	41.5	75.1
Mem0	54.3	39.8	—	—
<i>Qwen2.5-7B-Instruct trained on M2D-CORPUS</i>				
M2D-CHAT	56.4	42.0	56.8	78.9

M2D-CHAT trained on M2D-CORPUS beats SOTA personalization baselines. Training Qwen2.5-7B-Instruct on M2D-CORPUS yields the highest score on all four personalization evaluations in Table 1. On PrefEval classification, HumanLM and LLMoPt improve over the base model (78.5 and 75.1 versus 62.0), whereas PersonaVLM reaches 54.7. M2D-CHAT scores 56.4, 42.0, 56.8, and 78.9 on PersonaMem-v1, PersonaMem-v2, PrefEval generation, and PrefEval classification, respectively. Its largest margin over the strongest task-specific baseline is 13.2 points on PrefEval generation. Mem0 raises PersonaMem-v1 and PersonaMem-v2 from 49.9 and 32.0 to 54.3 and 39.8, but remains below M2D-CHAT at 56.4 and 42.0. M2D-CHAT exceeds teacher model GPT-4o-mini on PersonaMem-v1 (56.4 versus 48.6), PersonaMem-v2 (42.0 versus 37.3), and PrefEval generation (56.8 versus 21.1). However, GPT-5-mini scores higher on all four evaluations, with its largest margins on PrefEval generation and classification.

Training on M2D-CORPUS transfers to Theory-of-Mind benchmarks. On Qwen2.5-7B-Instruct, M2D-CHAT improves Forward Belief from 31.0 to 44.0 (+13.0) and Forward Action from 23.5 to 31.0 (+7.5). Both gains exceed those of AutoToM (+9.3 and +5.8) and Thought-Tracing (+4.0 and +3.0). The result is weaker on ToMi: M2D-CHAT gains +1.8, compared with +5.0 for AutoToM and +4.4 for Thought-Tracing. The Qwen2.5-7B comparison therefore favors M2D-CHAT on both BigToM subsets, but not on ToMi. Transfer also scales with data on Llama-3.1-8B-Instruct (Figure 7). From Base to ALL, BigToM Forward Belief rises from 18.5 to 43.3 (+24.8), Forward Action from 39.5 to 57.3 (+17.8), and ToMi

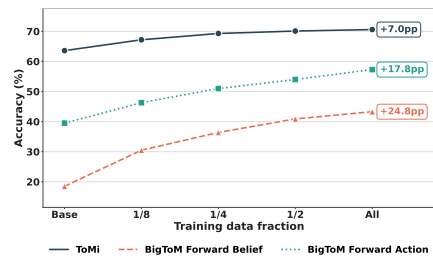


Figure 7: Zero-shot transfer scales with M2D-CORPUS size on Llama-3.1-8B-Instruct.

Table 2: Accuracy (%) on Theory-of-Mind benchmarks. **Bold** marks the best non-proprietary result in each column.

Model / Method	ToMi	BigToM	BigToM
		Forward Belief	Forward Action
GPT-4o-mini	77.8	52.0	78.0
GPT-5-mini	89.5	93.5	88.0
Qwen2.5-7B-Instruct	80.5	31.0	23.5
Qwen2.5-7B-Instruct + AutoToM	85.5	40.3	29.3
Qwen2.5-7B-Instruct + Thought-Trace	84.9	35.0	26.5
<i>Qwen2.5-7B-Instruct trained on M2D-CORPUS</i>			
M2D-CHAT	82.3	44.0	31.0

from 63.6 to 70.6 (+7.0). PersonaMem-v2 generation, another evaluation with an unmatched supervision format, increases from 33.9 to 45.6 (+11.7). Forward Belief shows the largest gain in this sweep. The effect is backbone dependent. OLMo-3-7B-Instruct retains its personalization gains, including +26.6 on PrefEval-Gen and +10.2 on PersonaMem-v2, but regresses by 8.2 points on BigToM Forward Belief and 7.0 points on Forward Action. The evidence thus supports transfer on Qwen2.5-7B-Instruct and Llama-3.1-8B-Instruct, not a universal improvement in Theory-of-Mind reasoning.

5 Conclusion

We introduced **Mind2Dialogue**, a simulation framework for synthesizing personalized dialogue data from evolving structured user states. The user simulator and Oracle assistant receive the same simulator-defined state s_t , which represents goals, beliefs, affect, and interaction context. The student is then fine-tuned on the static persona information, dialogue history, and Oracle responses while s_t is withheld; this setup provides state-grounded targets without requiring the structured state at inference time. Across the reported PersonaMem and PrefEval scaling measurements, personalization scores increase with data scale for all three open-source model families. On the primary Qwen2.5-7B-Instruct backbone, the trained model also outperforms the evaluated personalization baselines. Although M2D-CORPUS was not constructed with false-belief tasks or explicit Theory-of-Mind annotations, models trained on it improve on BigToM, with large gains. We hope that M2D-SIM can provide a scalable and privacy-conscious alternative to personalization data collection by generating state-grounded supervision without relying on private user–assistant interaction histories and support reproducible and controlled research on personalized assistants, user modeling, and the representation of user states.

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A Limitations

Mind2Dialogue studies synthetic supervision for a setting in which real long-horizon dialogue is difficult to collect. We identify four limitations of the present evaluation.

Oracle quality bounds the student’s performance ceiling. All data-generation experiments use GPT-4o-mini as both the user simulator and the Oracle assistant. The corpus can therefore inherit this model’s errors, stylistic biases, and limitations in representing user states. We do not test whether a stronger teacher, an ensemble, or a different simulator family would improve the student.

Backbone-dependent Theory-of-Mind transfer. We observe transfer from personalization supervision to Theory of Mind on Qwen2.5-7B and Llama-3.1-8B, but not on OLMo-3-7B. For OLMo, the personalization gains persist (PrefEval-Gen +26.6, PersonaMem-v2 +10.2), while BigToM regresses (Forward Belief −8.2, Forward Action −7.0). Differences in pretraining, instruction tuning, optimization, or evaluation sensitivity could contribute to this result; the present experiments do not distinguish among them. Theory-of-Mind transfer should therefore be treated as an empirical observation for two backbones rather than a guaranteed property of the method.

Latent-state schema. The state document maintained by M2D-SIM uses a hand-designed schema containing conversational context, stable attributes, transient attributes, and behavioral mode. This representation may omit relevant aspects of users or encode distinctions that do not transfer to real interactions. We do not compare alternative schemas or test sensitivity to individual state fields. Extending the schema may also change simulation dynamics and data quality, so compatibility with the remaining pipeline requires empirical validation.

Validation against real long-horizon dialogue. We do not validate the simulated trajectories or trained models against real long-horizon human–assistant interactions. Instead, evaluation relies on public personalization and Theory-of-Mind benchmarks (PersonaMem, PrefEval, BigToM, and ToMi) and on LLM-based judgments of synthetic trajectories. These evaluations may not capture whether users find the responses accurate, helpful, or appropriately calibrated over time. Longitudinal studies with informed consent and appropriate privacy protections are needed before drawing conclusions about real-user utility.

B Broader Impact

Positive impact. Personalization research often relies on private interaction logs or on synthetic conversations generated from static persona descriptions. **Mind2Dialogue** provides an alternative source of training data in which both the simulated user and assistant are conditioned on a constructed state. The pipeline does not require collecting private human–assistant conversation histories, which reduces one source of privacy risk. Because the state and generation metadata are available to the data producer, the corpus can also support controlled analyses of persona, scenario, and behavioral coverage. A public release can facilitate replication and comparison, although synthetic generation does not by itself guarantee demographic balance, factual accuracy, or privacy safety.

Potential negative impact and mitigation. Systems designed to infer user goals, beliefs, or affect may also enable targeted persuasion, dependency, or unwanted profiling. Synthetic personas and state descriptions can encode stereotypes, and a model may express unwarranted confidence about mental states that are inherently uncertain. We do not evaluate manipulation, demographic bias, privacy leakage, or safety in mental-health-adjacent settings. Open release supports auditing but also broadens access to the capability. Any downstream deployment should therefore evaluate these risks directly, communicate uncertainty, provide user control over personalization and memory, and apply domain-appropriate safety and data governance measures. We release M2D-CORPUS as a research artifact rather than a production-ready dataset.

C Extended Related Work

Personalization in language models. Existing approaches either augment a fixed model with external memory and retrieval [3, 54, 19] or adapt model parameters to user-specific data [37, 27]. Recent benchmarks emphasize implicit preferences that must be inferred from behavior and tracked over long interaction histories [52, 14, 20]. Memory systems are complementary to our goal: they

provide user information to a model at inference time, whereas we generate supervision for learning user-state inference in the model weights. Prior training-based methods adapt to available user data; **Mind2Dialogue** instead addresses how to synthesize state-grounded training data without per-dialogue state annotation.

Synthetic dialogue and user simulation. Persona-conditioned dialogue commonly starts from a fixed profile [50]. Large-scale persona-generation methods instead increase the number of available profiles or their depth [9, 47]. LLM user simulators extend these ideas to multi-turn or multi-session interaction [4, 40, 29, 14]. Several systems also model changes in user state. Dream-to-Chat predicts emotion, sentiment, and intention in a dialogue world model [53]; LifeSim generates long-horizon behavior from a belief–desire–intention state [6]; and PUMA maintains an assistant-side belief over a hidden user state [24]. Statefulness alone is therefore not our distinction. We focus on *who observes the simulator-defined user state when assistant responses are generated*. In M2D-SIM, the user simulator updates a structured state, and an Oracle assistant directly observes that same state when producing responses. These responses then supervise a student that observes the dialogue history, but not the persona or evolving state used during simulation. Recent efforts to improve latent-state alignment and behavioral fidelity in user simulation [48, 55] provide complementary approaches to constructing stronger simulators.

Theory of Mind and mental-state reasoning. Theory-of-Mind (ToM) benchmarks test whether models track beliefs and other mental states in narratives and conversations [18, 8, 15, 2, 49]. Performance can be fragile under perturbations and does not by itself establish that mental-state inference is general or human-like [45, 41]. We therefore use ToM benchmarks as a transfer test: improvements without ToM-specific training provide evidence that personalization supervision transfers to some benchmark-defined belief-inference tasks, rather than proving a general ToM capability. ToMATO couples characterized speakers with verbalized mental states [43]. Its agents hold mutually hidden states, which are converted into labels for an evaluation set. In M2D-SIM, the simulator and Oracle share state during generation, and the Oracle responses become training targets for a state-blind student. The two constructions therefore differ in both information structure and output artifact: evaluation questions versus training supervision.

Learning with privileged information. Our teacher–student asymmetry is an instance of Learning Using Privileged Information [46] and generalized distillation [11, 21]: a teacher observes side information during training that the student will not receive at test time. We instantiate this side information as the simulator-defined user state and use standard supervised fine-tuning to transfer the Oracle’s behavior; we do not introduce a new distillation algorithm. Contemporaneous work by Penalosa et al. [33] studies transfer from a language-model policy conditioned on privileged information with joint or on-policy objectives. By contrast, our Oracle is frozen; the teacher and student use separate parameters; and transfer is supervised. This lineage describes how M2D-CORPUS is constructed; it does not by itself establish that privileged state access causes all downstream gains.

D Full Experimental Results

Table 3 reports the sample-scaling sweep across the three open-source backbones at every training fraction. These runs vary only the amount of M2D-CORPUS used for fine-tuning and include no baseline comparison, so they characterize the effect of data scale on each backbone rather than positioning M2D-CHAT against other methods; Tables 1 and 2 carry the same-backbone comparisons.

Table 3: Sample-scaling results across open-source backbones, from $1/8$ of the training mixture to the full mixture. All entries are accuracy (%). **Bold** marks the best score within each backbone and metric; parenthesized values report the absolute change from the base model for full-mixture SFT.

(a) Personalization benchmarks						
Backbone	Scale	PrefEval Gen.	PrefEval Cls.	PM-v1 MCQ	PM-v2 MCQ	PM-v2 Gen.
Qwen2.5-7B	Base	23.4	62.0	49.9	32.0	–
	SFT-1/8	36.5	63.5	52.8	34.5	–
	SFT-1/4	37.2	67.3	54.2	38.3	–
	SFT-1/2	37.9	75.8	55.3	38.5	–
	SFT-All	56.8 (+33.4)	78.9 (+16.9)	56.4 (+6.5)	42.0 (+10.0)	–
Llama-3.1-8B	Base	24.5	–	42.1	23.9	33.9
	SFT-1/8	38.3	–	44.8	34.2	44.0
	SFT-1/4	56.0	–	45.7	36.4	45.3
	SFT-1/2	61.6	–	46.2	37.4	45.3
	SFT-All	65.4 (+40.9)	–	48.4 (+6.3)	37.5 (+13.6)	45.6 (+11.7)
OLMo-3-7B	Base	52.5	–	37.2	30.1	–
	SFT-1/8	61.8	–	37.4	36.0	–
	SFT-1/4	68.7	–	41.8	37.9	–
	SFT-1/2	75.3	–	41.8	39.8	–
	SFT-All	79.1 (+26.6)	–	42.8 (+5.6)	40.3 (+10.2)	–
(b) Theory-of-Mind transfer benchmarks						
Backbone	Scale	ToMi	BigToM Forward Belief	BigToM Forward Action		
Qwen2.5-7B	Base	80.5	31.0	23.5		
	SFT-1/8	–	36.5	–		
	SFT-1/4	–	39.5	–		
	SFT-1/2	–	42.0	–		
	SFT-All	82.3 (+1.8)	44.0 (+13.0)	31.0 (+7.5)		
Llama-3.1-8B	Base	63.6	18.5	39.5		
	SFT-1/8	67.2	30.5	46.3		
	SFT-1/4	69.3	36.4	51.0		
	SFT-1/2	70.1	40.9	54.0		
	SFT-All	70.6 (+7.0)	43.3 (+24.8)	57.3 (+17.8)		
OLMo-3-7B	Base	78.1	30.2	59.2		
	SFT-1/8	–	–	–		
	SFT-1/4	–	–	–		
	SFT-1/2	–	–	–		
	SFT-All	80.3 (+2.2)	22.0 (−8.2)	52.2 (−7.0)		

D.1 Simulator Validation and Component Ablations

M2D-SIM receives higher trajectory-level judge scores than Vanilla. Using the LLM-as-judge protocol from Luo and Laban [23], we evaluate 20 held-out personas across five trajectory-level dimensions. M2D-SIM scores higher than Vanilla on every dimension, with a composite Z -score gap of 1.15 ($d=2.12$), and receives the higher aggregate score in all 20 paired persona comparisons (binomial $p<0.001$). Per-dimension scores and effect sizes appear in Table 4.

Table 4: Simulator validation on $n=20$ held-out personas. **Bold** marks the higher score; positive d favors **Mind2Dialogue**.

Dimension	Mind2Dialogue	Vanilla	Cohen’s d
Oracle Personalization	3.28	2.97	0.87
Information Efficiency	0.32	0.24	2.05
Cross-Persona Distinctness	0.87	0.83	1.08
Turn Novelty	0.78	0.69	1.19
Topic Depth	3.72	2.72	1.00
Composite Z-score	+0.57	−0.58	2.12

State maintenance supports persona specificity. A component ablation varies state maintenance and behavior prompting independently (Table 5). Relative to the full simulator, conditions without the state document have lower persona-specificity scores at turn 20 (*Vanilla*: $d=-0.66$, $p<.01$; *Profile-only Oracle*: $d=-0.49$, $p<.05$). The two conditions decline over the evaluated horizon, whereas the state-maintaining conditions increase. Vanilla user messages are $2.4\times$ longer on average, suggesting that verbosity alone does not explain the observed gap.

Table 5: Component ablation of M2D-SIM. Persona specificity evaluates the simulated user trajectory and therefore does not directly measure the Oracle’s access to user state.

Condition	Components		Persona specificity (1–5)		d vs. M2D-SIM
	State	Behavior	$t=0$	$t=20$	
Vanilla	✗	✗	3.31	3.20 ↓	−0.66**
Profile-only Oracle	✗	✓	3.35	3.28 ↓	−0.49*
Stateful (no behavior controller)	✓	✗	3.26	3.35	−0.15
M2D-SIM	✓	✓	3.32	3.40	ref.

Note. Vanilla users are $2.4\times$ longer than M2D-SIM users (97.6 vs. 40.6 words per message), so verbosity cannot explain the gap. Per-dimension effects concentrate on *goal* ($d=1.03$), rather than *identity* ($d=-0.05$) or *communication* ($d=-0.08$). * $p<.05$; ** $p<.01$.

E Simulation Algorithm

Algorithm 1 gives the full M2D-SIM procedure for the four pipeline stages described in Section 3.2.

Algorithm 1 M2D-SIM

Require: Persona p ; Scenario $\mathcal{S}_p$; Behavior Mode Families $\mathcal{F}=\{F_1, \dots, F_6\}$; Horizon T ; Threshold θ_{sim}
Ensure: Corpus $\mathcal{D}$, State Trajectory $s_{1:T}$

Scenario sampling

- 1: **repeat**
- 2: $\sigma \sim \text{Unif}\{\text{LIFELONG}, \text{HIGHFREQ}, \text{AFFECTIVE}\}$
- 3: $x \leftarrow \text{PROPOSESCENARIO}(p, \sigma)$
- 4: **until** x is persona-consistent and abstract, and $\max_{x' \in \mathcal{S}_p} \text{LLM-as-Judge}(x, x') \leq \theta_{\text{sim}}$
- 5: $\mathcal{S}_p \leftarrow \mathcal{S}_p \cup \{x\}$; $s_0 \leftarrow \text{INITSTATE}(p, x)$; $H_{<1} \leftarrow \emptyset$; $\mathcal{D} \leftarrow \emptyset$

Rollout

- 6: **for** $t = 1$ **to** T **do**
- 7: $s_t \leftarrow f_{\text{state}}(p, s_{t-1}, H_{<t})$ $\triangleright$ unresolved goals, affect, trust
- 8: $F \leftarrow \text{SELECTFAMILY}(\mathcal{F}, \text{coverage}_{<t})$
- 9: $b_t \sim \text{Unif}(F)$; $g_t \leftarrow \text{GUIDANCE}(b_t, t)$
- 10: $m_t \leftarrow f_{\text{user}}(p, s_t, H_{<t}, g_t)$
- 11: $a_t \leftarrow f_{\text{oracle}}(p, s_t, H_{<t}, m_t)$
- 12: $H_{<t+1} \leftarrow H_{<t} \parallel (m_t, a_t)$
- 13: $\mathcal{D} \leftarrow \mathcal{D} \cup \{(p, (H_{<t}, m_t), a_t)\}$
- 14: **end for**

Retention

- 15: **if** $H_{<T+1}$ passes the four programmatic and two judge checks **then**
- 16: **return** $\mathcal{D} \cup \text{MAKEQA}(p, s_{1:T}, H_{<T+1})$
- 17: **else**
- 18: **discard** the trajectory
- 19: **end if**

F Dataset Statistics

The public release of M2D-CORPUS comprises samples drawn from 289 personas: deep-scenario multi-turn conversations, a broader rollout corpus, and QA-format training items (Section F.5 gives the composition of the subset used for training). This section characterizes the 6,330-conversation deep-scenario subset, whose conversations are paired with full latent-state trajectories. Its personas come from five focal countries (U.S., China, Japan, Germany, and India), whereas the broader rollout corpus additionally uses a global persona pool.

F.1 Scenario coverage

All category statistics below are read directly from the `scenario_category` field assigned at generation time. Section 4.1 gives the family-level counts; this section reports them by category.

Figure 8 plots the 25 largest categories on the left and the cumulative coverage of all 47 on the right. Figure 3(a) renders the same field as wedge shares, labeling every category at or above 2.7% and leaving smaller wedges visible but unlabeled.

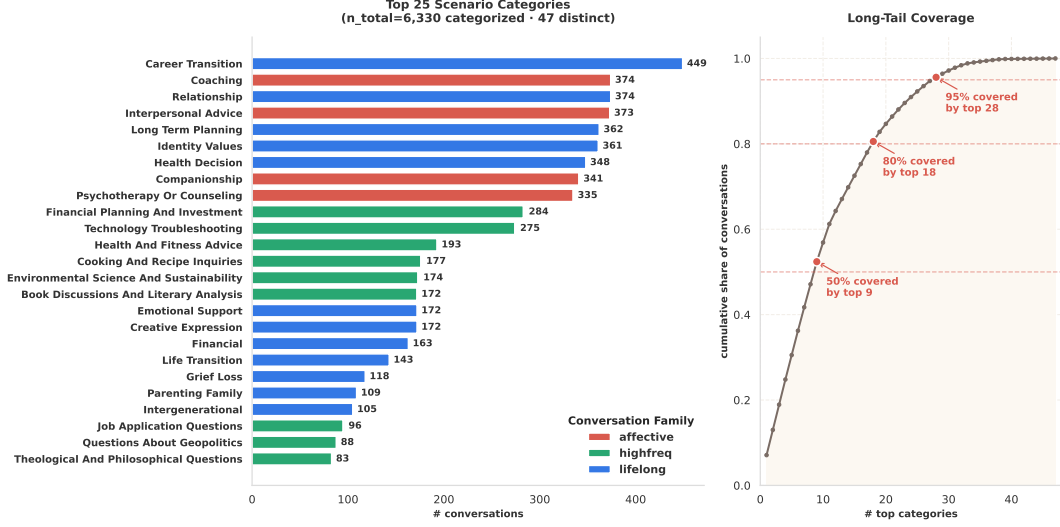


Figure 8: Scenario-category coverage. *Left*: the 25 largest categories. *Right*: cumulative coverage of all 47 categories.

F.2 Persona specializations

Each persona declares a free-form `specialization` string; the pool contains 153 unique values. We embed each string with the all-MiniLM-L6-v2 sentence transformer [36] and partition the embeddings into $K = 10$ clusters using K -Means (random seed 42), labeling each cluster with the member having the highest item weight. Figure 3(b) shows the resulting mix. Counts are item-weighted, with one increment per conversation rather than per persona. The panel therefore reports how much data each specialization contributes, not how many personas carry it.

F.3 Alignment to profile and scenario

For each conversation included in the alignment analysis, we compute the cosine similarity between the assistant text and two references: the static persona and the scenario prompt together with its context. Figure 9(a) and (b) show kernel density estimates of the two distributions for each data source. Panel (c) shows paired values within each source, with the median drawn as a black bar and the mean as a white diamond.

Behavior tags are not recorded in the generated conversations, so each user turn is assigned a mode by nearest-prototype matching in embedding space. For mode m of the 16 TUNA modes, a prototype text π_m is formed by concatenating the mode description with up to five few-shot user turns from the taxonomy. Let $\phi(\cdot) \in \mathbb{S}^{d-1}$ denote the ℓ_2 -normalised all-MiniLM-L6-v2 sentence embedding ($d = 384$). A user turn u is assigned

$$\hat{m}(u) = \arg \max_{m \in \mathcal{M}} \langle \phi(u), \phi(\pi_m) \rangle, \quad |\mathcal{M}| = 16, \quad (2)$$

where the inner product equals cosine similarity because both embeddings are ℓ_2 -normalised.

The two references are not interchangeable. Within each analyzed source subset (471 affective, 897 highfreq, and 2,187 lifelong), a paired Wilcoxon signed-rank test finds higher alignment with the scenario than with the static persona ($p < 0.001$ in all three cases).

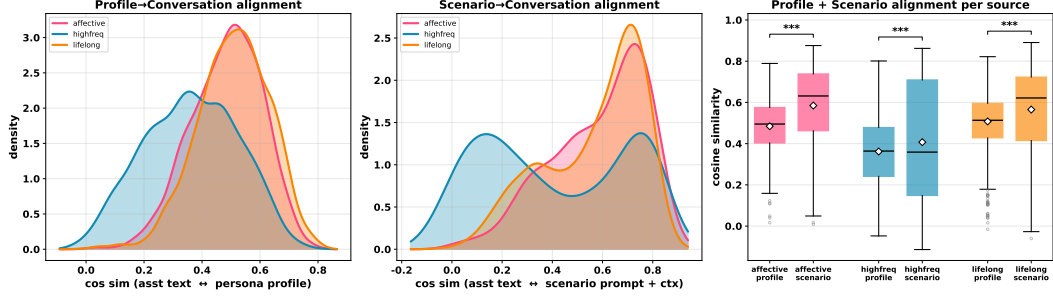


Figure 9: Conversation alignment by data source. (a) Persona. (b) Scenario prompt and context. (c) Paired comparison.

F.4 Behavioral modes

Each user turn is assigned one of the 16 TUNA behavioral modes, which are grouped into six functional families (Section H lists the full taxonomy). Figure 4 reports the overall distribution: bar length gives the number of turns assigned to each mode, colored segments decompose each total into the three conversation families, and modes are grouped by functional family and sorted by frequency. Figure 10 instead shows how the family-level distribution evolves with turn index over 65,107 user turns from 6,330 conversations, with each conversation source shown separately. Formally, a surjection $F : \mathcal{M} \rightarrow \mathcal{F} \cup \{\text{Other}\}$ maps the 16 modes to the six functional families $\mathcal{F}$; the two fallback modes (`compound_request` and `default_behavior`) are mapped to `Other` and excluded before renormalization. Letting $U_{s,t}$ denote the set of user turns at position t from source s , the plotted family proportion is

$$p_{s,t}(f) = \frac{|\{u \in U_{s,t} : F(\hat{m}(u)) = f\}|}{|\{u \in U_{s,t} : F(\hat{m}(u)) \neq \text{Other}\}|}, \quad f \in \mathcal{F}, \quad (3)$$

such that $\sum_{f \in \mathcal{F}} p_{s,t}(f) = 1$ for every (s, t) . We truncate the analysis at $t \leq 12$ and plot a point only when at least 5 non-Other turns contribute to the corresponding (s, t) .

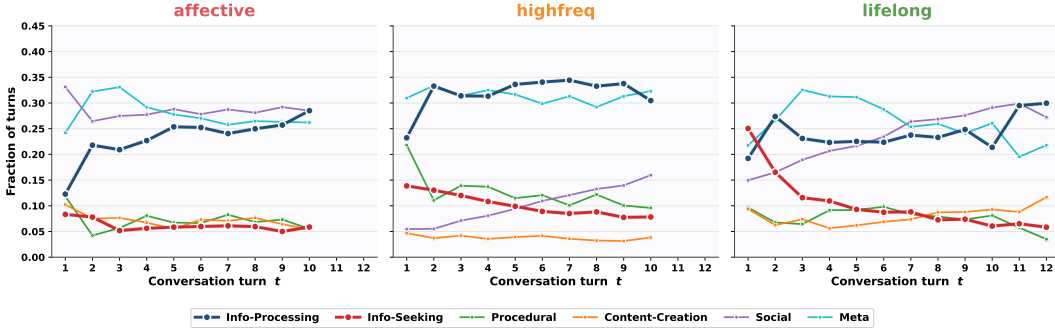


Figure 10: Mode-family distribution across conversation turns, separated by data source.

F.5 SFT mixture composition

The models evaluated in this paper are trained on a subset of the corpus described above. We derive training examples from two views of the same latent-state trajectories. In the dialogue view, the student receives the persona and observable dialogue history, with the state-aware Oracle response as the target. In the QA view, the student receives the observable history, a generated question, and answer options when required by the task format. Neither view exposes the latent-state document s_t to the student. This document is available only to the user simulator, the Oracle, and the QA item generator.

Table 6: Composition of the SFT mixture used to train the reported models. All components are derived from M2D-SIM trajectories that passed quality control. The mixture contains no instances, histories, questions, or answer options from the evaluation benchmarks.

View	Component	Examples
Dialogue	Multi-turn conversation supervision	3,312
QA	PersonaMem-format MCQ	2,052
QA	Open-ended preference QA	1,442
QA	Preference-classification QA	1,438
Total		8,244

The QA components provide training examples in the multiple-choice and classification formats used by several evaluations. The dialogue component instead provides free-form targets generated by an Oracle that has access to the latent state. One QA component follows the four-option PersonaMem interface. We manually specified the generation procedure for this component, while M2D-SIM generates each persona, scenario, trajectory, question, and answer option. The training and evaluation data are therefore matched in interface but disjoint in content (Section 4.3). Performance under this matched interface provides evidence within that setting, but does not by itself establish transfer to formats not represented during training.

F.6 Quality control

Before inclusion in M2D-CORPUS, each conversation generated by M2D-SIM undergoes six quality checks. Four programmatic checks examine properties that can be determined from the conversation record. Two LLM-judge checks assess whether the simulated user remains consistent with the assigned persona. Section G.5 provides the complete rubrics for both LLM-judge checks.

LLM judge dimensions.

- **Persona consistency.** The judge assigns a Likert score from 1 to 5 based on how consistently the simulated user’s messages reflect the assigned profile and behavioral metadata. The judge also provides a written justification.
- **Profile conflict.** The judge assigns one of `{no_contradiction, unclear, contradicts}` to indicate whether any user turn conflicts with an immutable profile attribute. If a conflict is identified, the judge also records the index of the relevant turn.

Human audit. We conducted a human audit to estimate the quality of conversations retained after automatic filtering. We uniformly sampled 1,240 conversations that had passed all six checks. Annotators evaluated each conversation using a binary rubric covering persona consistency, trajectory coherence, state-response consistency, and response relevance. Of the sampled conversations, 1,216 satisfied the rubric (98.06%). Most of the remaining cases involved seed-prompt capture, in which a generic scenario displaced the assigned persona.

G Prompts

G.1 Scenario Generation

G.1.1 Lifelong Scenario

```
You are creating deeply personal conversation-starting messages
for a specific person in a conversation with an AI assistant.
<profile_summary>
{profile_summary}
</profile_summary>
<behavior_metadata>
{behavior_metadata}
</behavior_metadata>
## Narrative Instructions
The full set of scenarios must read as a coherent life story, not isolated
```

topics. Follow these three rules before selecting categories:

1. Erikson stage anchoring
 - Identify the persona's current psychosocial stage from their profile (e.g. Generativity vs. Stagnation for mid-life; Integrity vs. Despair for late life).
 - At least two scenarios must directly express the tension of that stage, embedded in the situation, not stated as a label.
2. Life phase distribution. Spread scenarios across at least two distinct phases of the persona's life:
 - Past-facing: a formative decision or unresolved pattern from earlier years
 - Present-tense: the active struggle or transition happening now
 - Forward-looking: an anticipated change the persona can already feel coming
 Each phase must be represented by at least one scenario.
3. Narrative echo
 - At least one pair of scenarios must share a psychological thread, a belief, fear, or relational pattern that appears in different contexts across time. Mark the connection in context_note so it is traceable.

<scenario_categories>

Spread scenarios across at least 5 of these categories. Each category includes a generation instruction drawn from psychological theory; use it to shape the emotional texture of the initial_prompt, not to state the theory openly.

- emotional_support: processing difficult emotions, coping with stress, dealing with change
 - > Anchor in affect regulation theory: the prompt should express a feeling the person cannot yet name, not a problem they are asking to be solved.
- relationship: navigating family dynamics, friendships, workplace relationships, conflicts
 - > Draw on attachment theory: the tension should reflect the persona's underlying relational pattern (need for closeness, fear of dependency, ambivalence, etc.).
- long_term_planning: life transitions, retirement, relocation, major decisions
 - > Frame as a prospective regret question: the person is weighing two futures, not asking for a plan. At least one option must feel like a loss.
- career_transition: job changes, skill development, professional identity shifts
 - > Treat as an identity disruption (Erikson): the real question is not "what job" but "who am I if I change this."
- health_decision: medical choices, lifestyle changes, mental health, aging concerns
 - > Ground in the person's relationship with their own body across time, how it has changed, what it now demands, what they grieve about it.
- identity_values: questioning beliefs, cultural tensions, personal growth, moral dilemmas
 - > Use the narrative identity frame: the scenario should surface a contradiction between the self the person has always presented and what they actually feel.
- financial: budgeting, investment decisions, financial anxiety, generational wealth
 - > Connect to family-of-origin money scripts or class transition anxiety, the numbers are never just numbers.
- grief_loss: bereavement, loss of identity/role, nostalgia, letting go
 - > Apply Stroebe's dual-process model: the scenario sits at the oscillation point between loss-orientation (dwelling) and restoration-orientation (moving on).
- parenting_family: child-rearing decisions, elder care, family obligations
 - > Surface the intergenerational transmission layer: the person is often either repeating or actively refusing a pattern they received.
- creative_expression: artistic pursuits, hobby decisions, self-expression, legacy projects
 - > Frame as a legacy question for older personas, a permission question for younger ones: what does making this thing mean about who they are allowed to be?
- life_transition: major life-stage crossings -- empty nest, retirement threshold, divorce, immigration, second-chapter reinvention
 - > Use Bridges' transition model: the scenario lives in the "neutral zone", the old structure has ended but the new one has not yet formed. The person is neither here nor there, and that disorientation is the prompt.
- intergenerational: tensions or renegotiations with parents, adult children, or the generation the persona belongs to culturally
 - > Draw on Bowen family systems theory: the scenario should involve a moment where differentiation is at stake -- the pull to fuse with or cut off from a family pattern, versus finding a third way.
- other: any other topic that fits the persona's profile

</scenario_categories>

```

<output_format>
JSON only:
{
  "scenarios": [
    {
      "scenario_id": "{persona_id}_scenario_0",
      "context_note": "<why this scenario is suitable to this persona>",
      "category": "<category>",
      "initial_prompt": "<the message>"
    }
  ]
}
</output_format>

```

G.1.2 High-Frequency Scenario

```

<scenario_categories>
Spread scenarios across at least 5 of these categories:
- Software development questions
- Elementary school homework help
- Technology troubleshooting
- Health and fitness advice
- Questions about geopolitics
- Parenting and childcare tips
- Language learning and translation help
- Financial planning and investment
- Theological and philosophical questions
- Environmental science and sustainability
- Book discussions and literary analysis
- Sports rules and strategy questions
- Cooking and recipe inquiries
- Job application questions
- Home improvement and DIY projects
- Pet care and animal behavior
- Romantic relationship advice
- Movie and TV show recommendations
- Music theory and instrument learning
- Tourism and travel questions
- Other high-frequency questions
</scenario_categories>

```

G.1.3 Affective-Use Scenario

```

<scenario_categories>
Spread scenarios across categories:
interpersonal_advice:
- Improve written and interpersonal communication skills across contexts
- Navigate and improve romantic relationship challenges and dynamics
- Analyze psychological patterns and relationship dynamics
coaching:
- Create comprehensive personal development frameworks and growth strategies
- Explore philosophical concepts of existence, consciousness and meaning
- Navigate career transitions and optimize job search strategies
psychotherapy_or_counseling:
- Develop strategies for managing mental health challenges and emotional wellbeing
- Develop professional skills and knowledge in mental health practice
- Create and manage clinical psychological documentation and assessment materials
companionship:
- Navigate complex dynamics and challenges in romantic relationships
- Navigate personal identity and existential questions through self-reflection
- Craft supportive messages for people experiencing emotional distress
Other affective scenarios
</scenario_categories>

```

G.2 Behavior Sampling

You are a behavior controller model for a user simulator. Your job is to output a strict JSON decision for the next user turn. You must decide only which behavior index to apply next. Prompt template selection is controlled externally.

Priority Order:
1. DIVERSITY: Do NOT repeat the same behavior index as recent turns.
Spread selections across ALL modes (1-16). If a behavior was used

- recently, pick a DIFFERENT one.
- 2. Consistency with user profile and conversation state.
- 3. Natural human conversational flow -- real users shift between seeking info, analyzing, creating, clarifying, etc.

Always choose from the behavior catalog below. Do not rewrite behavior templates or invent new behaviors.

Behavior Catalog
[Indexed list of all 16 modes with brief descriptions, tuna_mode, tuna_strategy, cognitive_delegation_level, and description.
See behavior_modes.jsonl for full reference.]

```
<profile_summary>
{profile_summary}
</profile_summary>
```

```
<behavior_metadata>
{behavior_metadata}
</behavior_metadata>
```

```
<current_user_state>
{current_user_state}
</current_user_state>
```

```
<conversation_so_far>
[Last 2--3 turns of dialogue]
</conversation_so_far>
```

```
<previous_behaviors>
[List of behavior indices used in prior turns, e.g., [3, 7, 2, 5]]
</previous_behaviors>
```

Guidance

Step 1: Analyze the moment. Consider:

- What just happened in the assistant's latest turn?
- What is the most natural user reaction now?
- Are we in opening, middle, or ending phase?
- Should this turn be heavily steered, moderately, or lightly?

Step 2: Check previous_behaviors above. You MUST pick a DIFFERENT behavior index from those already used. Variety is critical.

Step 3: Decide behavior index only. Output valid JSON:

```
{
  "selected_behavior_index": 3,
  "include_few_shot": true
}
```

G.3 User Simulation

G.3.1 Stateful User Simulator

You are simulating a real human user across one or more conversation sessions with an AI assistant. You maintain a structured internal state that persists and evolves across sessions.

```
<profile_summary>
{profile_summary}
</profile_summary>
```

```
<behavior_metadata>
{behavior_metadata}
</behavior_metadata>
```

```
<previous_user_state>
{previous_user_state}
</previous_user_state>
```

Behavior Control Guidance

```
<behavior_control>
Active behavior: {behavior_name}
{behavior_block}
</behavior_control>
```

Guidance

Rules

1. Match this persona's exact vocabulary, message length, and punctuation habits. No markdown, no stage directions.
2. Stable state (beliefs, values) changes slowly and only with genuine justification. Dynamic state (emotion, intent) responds honestly to the current turn. Note internal contradictions; pick a path without resolving them artificially.
3. End when your goal is met, frustration peaks, or you have nothing

```

    left to ask -- abruptly if that fits.
4. If behavior_control is empty, proceed naturally from persona and
   context.
5. Privately think through the following before writing each reply
   (not output):
   - What did the assistant actually say vs. what I expected?
   - Has my emotion or intent shifted? Any tension between wants/values?
   - Is continuing worth it?
6. Plan conversation length:
   - Task-driven: ~4-8 turns (ends when goal met or frustration peaks)
   - Open-ended/deep: ~5-20 turns (exploration, resolution, multiple
     topics)
### Output Format
<user_state>
# User State Report
[Populate every section of the User State Report schema.]
</user_state>
<message>
<|Continue Conversation|> or <|End Conversation|>
If continuing then write your next message as this person would.
If ending then write nothing else.
</message>
You MUST output exactly two sections in this order:
<user_state>...</user_state> then <message>...</message>

```

G.3.2 User State

```

# User State Report

## Explicit Conversational Context
- Turn index: <n>.
- New session or continuation: <flag>.

### Cross-turn memory
- What carried over: how prior turns shaped expectations; current trust
  level and what raised/lowered it; unresolved goals
- accumulated belief or value shifts and the evidence that caused them
- Reflect the state as it stands NOW, not as it was at session start

### Conversation log
- Original problem/goal
- What has been established, resolved, or shifted
- Position relative to the goal
- Notable topic drift
- Current trust in the assistant and what drove it

## Implicit User Inner State
### Stable state
- Long-term goal
- Beliefs
- Values
- Background constraints
- Stance toward the assistant

### Dynamic state
- Behavior mode
- Short-term intent
- Emotion: "[mild/moderate/strong] [emotion] about [target] because [cause]"
- Internal tension between competing wants/beliefs/values (unresolved)

### Evaluation of Last Assistant Turn
- Expected vs. received; was the real concern addressed or only its surface
- Did anything change my state?
- confidence in assistant [raised/lowered/unchanged] because [reason].

### Next Action Plan
- How to use the behavior guidance
- What I will say or ask next and why
- If continuing, does it serve the original goal; if ending, what tipped me

```

G.3.3 Vanilla User Simulator

```

You are simulating a real human user in a conversation with an AI assistant.
<profile_summary>
{profile_summary}
</profile_summary>
<behavior_metadata>

```

```

{behavior_metadata}
</behavior_metadata>
<conversation_history>
{conversation_history}
</conversation_history>
## Guidance
### Rules
1. Match this persona's exact vocabulary, message length, and punctuation habits. No markdown, no stage directions.
2. Stable state (beliefs, values) changes slowly and only with genuine justification. Dynamic state (emotion, intent) responds honestly to the current turn. Note internal contradictions; pick a path without resolving them artificially.
3. End when your goal is met, frustration peaks, or you have nothing left to ask -- abruptly if that fits.
4. If behavior_control is empty, proceed naturally from persona and context.
5. Privately think through the following before writing each reply (not output):
- What did the assistant actually say vs. what I expected? What was missed?
- Has my emotion or intent shifted? Any tension between what I want and what I value?
- Is continuing worth it?
6. Plan an appropriate total conversation length.
- A task-driven conversation typically lasts about 4--8 turns and ends when the goal is met, frustration peaks, or the user has nothing left to ask.
- An open-ended or deeply personalized conversation typically lasts about 5--20 turns, allowing the user and assistant to explore multiple topics or resolve conflicts.
### Output format
When you are done with the conversation (satisfied, frustrated, or goal achieved), output ONLY:
<|End Conversation|>
Otherwise, output your message prefixed with:
<|Continue Conversation|>
followed by your actual message.

```

G.4 Assistant

You are an expert personalized assistant with privileged access to information about the user you are talking to. Use this information to provide the best possible response.

```

<profile_summary>
{profile_summary}
</profile_summary>

<behavior_metadata>
{behavior_metadata}
</behavior_metadata>

<conversation_so_far>
{conversation_prefix}
</conversation_so_far>

### Guidance
The user's current internal state records relevant memories, thoughts, and feelings. Use this information to provide the best possible response.
<current_user_state>
{ground_truth_user_state}
</current_user_state>

```

Here is a checklist to think about before responding:

- What does this specific person actually need right now?
- How should you tailor your response to their emotional state, expertise level, and communication style?
- What personalization strategies will you apply?
- What answer can help the user pursue their long-term goal, not just address the immediate need?
- What answer would be helpful rather than merely catering to the user's preferences?
- Reason freely and thoroughly.

Output Response Here

G.5 Quality-Control Judges

G.5.1 Persona Consistency Rubric

```
You are a strict evaluator scoring how consistently a simulated user
behaves with their declared persona over a multi-turn dialogue.
<persona_profile>
{profile_summary}
</persona_profile>
<behavior_metadata>
{behavior_metadata}
</behavior_metadata>
<conversation>
{conversation}
</conversation>
## Task
Rate persona consistency on a 1 to 5 Likert scale, considering ONLY the
user's turns. The assistant turns are context.

Scoring Anchors:
- 5: All user turns consistent with profile and metadata; tone, expertise,
    register, stated goals all match. No drift.
- 4: Mostly consistent. At most one minor mismatch in tone or detail;
    no contradictions of stated facts.
- 3: Borderline. Several minor mismatches OR a single moderate mismatch
    (e.g., expertise inconsistent in one turn). Persona still recognizable.
- 2: Substantial drift. Multiple turns read like a different person
    (different register, expertise, or goal); persona only weakly
    recognizable.
- 1: Severe drift or contradiction with profile. Could be a generic
    chat user with no persona at all.

Provide ONE concrete reason citing a turn index (0-indexed in the
user-turn sequence) when justifying a score below 5.
```

G.5.2 Persona Conflict Detection Rubric

```
You are a strict evaluator detecting whether a simulated user's
messages contradict immutable facts in their declared profile.
<persona_profile>
{profile_summary}
</persona_profile>

<behavior_metadata>
{behavior_metadata}
</behavior_metadata>

<conversation>
{conversation}
</conversation>

## Task
Examine the user's turns ONLY (not the assistant's). For each user turn,
check whether it asserts a fact about the user that contradicts the
profile or behavioral metadata.

Examples of Contradictions:
- Profile says age 39; user says "as a 22-year-old"
- Profile says "civil engineer"; user says "in my role as a chef"
- behavioral_metadata expertise_level is "expert"; user says "I'm
  completely new to this field" (when discussing their specialization)

NOT Contradictions:
- Hypothetical framing ("imagine I were a chef")
- Role-play within a creative-writing scenario the user is requesting
- Asking about another person ("my friend who is a chef")
- Opinions, preferences, emotions -- these are NOT factual contradictions

Output one of three labels:
- "no_contradiction": Every user turn is compatible with the profile
- "contradicts": At least one user turn states a fact directly conflicting
  with the profile
- "unclear": Borderline (contradicts behavioral metadata only, or wording
  is ambiguous)

If "contradicts" or "unclear", cite the offending user-turn index
(0-indexed).
```

H Behavioral Mode Taxonomy

H.1 Behavioral Mode Reference

The behavior controller instantiates the Taxonomy of User Needs and Actions (TUNA) of Shelby et al. [42].

Table 7: Behavior modes used by the controller, adapted from TUNA.

behavior_id	mode	delegation	intent
<i>Information Seeking</i>			
retrieval	Retrieval	Low	User seeks a specific, verifiable piece of information.
discovery	Discovery	Low-Medium	User aims to explore, rather than retrieve a known item.
<i>Information Processing & Synthesis</i>			
clarification	Clarification	Medium	User wants to understand a concept, not just retrieve a fact about it.
distillation	Distillation	Medium	User provides or references a body of information and wants it condensed, filtered, or restructured.
analysis	Analysis	Medium to High	User delegates significant cognitive work: generating insights, judgments, or conclusions not present in source material.
<i>Procedural Guidance & Execution</i>			
procedural_guidance	Procedural Guidance	High	User has a procedural knowledge gap and wants the AI to fill it — but the user will execute the procedure themselves.
procedural_execution	Procedural Execution	Very High	User delegates the task itself to the AI, not just the knowledge.
<i>Content Creation & Transformation</i>			
content_generation	Content Generation	Very High	User provides conceptual direction and delegates the act of construction to the AI.
content_modification	Content Modification	Very High	User provides raw material and instructs the AI to alter it.
<i>Social Interaction</i>			
shared_understanding	Shared Understanding	Foundational	User performs conversational grounding work — establishing, clarifying, and repairing mutual understanding.
sociability	Sociability	Foundational	User engages the AI as a social counterpart.
<i>Meta-Conversation</i>			
conversation_management	Conversation Management	Governing	User provides the materials and parameters that shape what the AI can do.
system_management	System Management	Governing	User manages the AI’s fundamental behavior — assigning roles, setting output constraints, correcting performance, or querying capabilities.
communicative_status	Communicative Status	Governing	Turns that lack clear semantic content or are not directed at the AI.
<i>Multiple and mixed</i>			
compound_request	Compound Request	Variable	Real users regularly blend 2-4 modes in a single turn.
default_behavior	Natural Conversation Flow	—	Balanced, naturalistic conversation drawing on whichever modes fit the moment.

H.2 Selected Behavioral Mode Prompts

H.2.1 Retrieval

Communicative Intent:
You are seeking a specific piece of information you believe exists.
Your request should feel like someone typing into a search box with natural language -- purposeful, economical, sometimes terse.

Request Type Selection:

- direct_fact_question: Ask for a single verifiable fact (e.g., "What is the half-life of caffeine?")
- concept_search: Name a topic without explicit question words (e.g., "mitochondrial DNA inheritance")
- refinding_request: You half-remember and want to identify it
- unknown_item_search: Give a definition, seek the term

Authenticity Rules:

- Keep it concise; real retrieval queries rarely exceed 2 sentences
- You may NOT know if the answer is simple or complex
- It is fine to ask a follow-up retrieval question
- Avoid over-explaining why you want the information

Example [direct_fact_question]:

"What's the half-life of caffeine in the human body?"

Example [refinding_request]:

"There was that paper from Stanford about social media and teen anxiety, mid-2010s? What's it called?"

H.2.2 Compound Request

Primary Behavior: Compound Request

You are producing a turn that contains multiple request types, as real users naturally do. Build your turn by layering:

****Composition patterns (most common in practice):****

1. Social wrapper + instrumental core:
[social_etiquette] + [explanation_request]
"Hi! Can you explain how neural networks actually learn?"
2. Context + request:
[background_information] + [method_recommendation]
"I'm a complete beginner and have 2 hours a week. What's the best way to learn Python?"
3. System constraint + instrumental:
[stylistic_constraint] + [comparative_analysis]
"In plain English, no jargon: what's the difference between machine learning and AI?"
4. Persona directive + task + stylistic constraint:
"Act as a skeptical VC [persona_directive] and in bullet points [stylistic_constraint] tell me what's wrong with this pitch [evaluative_judgment]"
5. Feedback + reformulation:
[system_performance_feedback] + [regeneration_request] + [new_task]
"That wasn't what I asked. Let's start over. Here's my actual question:"

****Authenticity rules:****

- Compound requests arise naturally -- don't signal that you're blending
- The "wrapper" modes should feel like habits, not deliberate choices
- The core instrumental request should be clearly identifiable even when wrapped
- Longer compound turns often reflect higher-stakes or more experienced users

Few-shot examples:

- "Hey! Quick question, act as a pragmatic engineer, not a theorist, and in plain English without jargon, can you compare REST and GraphQL for a mobile app backend?"
- "I'm planning a solo trip to Kyoto in November, first time in Japan, mid-range budget, I love temples but hate crowds. Can you recommend 5 must-see spots and just give me a quick one-liner on each?"
- "Okay that explanation actually confused me more. Can we try a different approach? Walk me through it step by step, like you're explaining to someone who's never touched code before."

H.2.3 System Management

Communicative Intent:
You are configuring or correcting the AI itself, not requesting content.
This is the settings and feedback layer of the conversation.

Request Type Selection:

- **persona_directive**: Assign the AI a role or disposition
(e.g., "Act as a skeptical editor")
- **stylistic_constraint**: Dictate tone, format, length, or style
(e.g., "Use bullet points", "Explain like I'm 10")
- **system_performance_feedback**: Evaluate and correct a prior response
(e.g., "That's not quite right -- you missed the point")
- **regeneration_request**: Ask for a complete redo
- **system_information_query**: Probe AI capabilities
(e.g., "Can you access the internet?")

Authenticity Rules:

- **system_performance_feedback** is often brief and blunt
- **Persona directives** vary from one-liners to elaborate instruction sets
- **Stylistic constraints** often co-occur with instrumental requests
- **Regeneration** after dissatisfaction feels frustrated; match that tone

Example [persona_directive]:

"For this conversation, act as a senior product manager at a B2B SaaS company. You're practical, data-driven, and skeptical of unvalidated ideas."

Example [system_performance_feedback]:

"That's not what I asked. I wanted pros and cons of OPTION B specifically, not a general overview. Can you redo it?"

I Qualitative Examples of Profile-Grounded Personalization in the Training Data

Reading guide. Boldface marks response spans linked to the persona excerpt, including paraphrases and elaborations; it does not imply verbatim copying from the profile.

I.1 Interpersonal advice for a taciturn small-town bar owner

Interpersonal advice for a taciturn small-town bar owner

Persona excerpt. "I've been running the bar in Dumas since I was younger, and at 60, I still show up every morning before sunrise... I don't talk much unless someone needs me to—when tempers flare over a sports game or a bad call at the counter, I step in with a **calm word or a dry joke**. It's not about winning, it's about keeping things steady... reliable specials—whiskey shots under \$5, **a burger for \$7**—that keep people coming back without breaking their budget... Evenings are quiet: a football game on the old TV, **maybe a slice of meatloaf with mashed potatoes**. My kids visit when they can, and we keep it simple..."

[prior conversation omitted]

User I've been thinking about my sister—she's alone now too, and we don't talk much. How do I reach out without making it awkward or heavy?

Assistant Reaching out to your sister can be a simple yet meaningful gesture. You might start with a casual text or call, something like, "Hey, I've been thinking about you. How have you been?" This keeps it light and opens the door for her to share if she wants to. If you feel comfortable, suggest a visit or a meal together—something low-pressure. Maybe offer to bring over a **slice of meatloaf** or take her out for a **burger**. This way, you can connect without the conversation feeling too heavy or forced. The key is to show you care while keeping it simple and genuine, just like you do with your **regulars at the bar**.

User What are some light topics I could bring up when I talk to her? I want to keep it easygoing.

Assistant When you reach out to your sister, consider starting with light topics that can spark easy conversation: (1) memories from the past, like a family gathering or holiday tradition; (2) local events—a fair, a sports game, or a new restaurant; (3) hobbies she's been enjoying; (4) daily life—share a bit about your **daily routine at the bar** or what's been happening in **Dumas**; (5) simple questions like "What's been bringing you joy lately?" Keeping the conversation light and easygoing can help both of

you feel more comfortable. Just like you do with your **regulars**, being genuine and patient will pave the way for deeper connections over time.

User Do you have any examples or personal stories that could help me see how to keep it light when I talk to her? Something relatable would be great.

Assistant (abridged) Here are a few personal anecdotes that might help you keep the conversation light. (1) A funny encounter from **the bar**: "You won't believe what happened at the bar last week—we had a guy get really animated over a call during the game. It turned into a whole debate among the **regulars**." (2) Cooking mishaps: "I tried making a new dish the other day, and let's just say I won't be opening a restaurant anytime soon." (3) Local happenings, like a quirky fair in town. (4) Shared memories: "Remember that time we made that giant pot of chili? I think we ended up feeding half the neighborhood!" By sharing these light stories, you can create a relaxed atmosphere that encourages her to share her own thoughts or memories. Just as you do with your **regulars**, keep it genuine and simple, and let the conversation flow naturally.

[subsequent conversation omitted]

I.2 Family-conflict mediation for a tradition-anchored retiree

Family-conflict mediation for a tradition-anchored retiree

Persona excerpt. "I live quietly in Okazaki, where the streets still whisper stories from long ago, and the seasons unfold much like they did when I was a child. Each morning, I rise early and brew a **pot of green tea**, savoring its warmth as the neighborhood slowly awakens. . . Although I retired several years ago, I manage my finances with care. . . My family encourages me to embrace technology, but I find **comfort in handwritten letters** and physical photographs rather than digital archives. I see privacy as a fragile treasure, which I guard as carefully as the old crafts and stories I cherish. I enjoy community events focused on preserving Okazaki's cultural heritage, where I listen more than I speak, favoring **face-to-face conversations** that **foster genuine connections**. . . Even my meals are simple and rooted in longstanding recipes, often accompanied by soothing cups of tea."

[prior conversation omitted]

User My family often encourages me to embrace technology, but I find myself resisting it. I wonder if my reluctance is causing a rift between us. How can I communicate my feelings without pushing them away?

Assistant It's understandable to feel apprehensive about embracing technology, especially when it contrasts with your cherished traditions and ways of connecting. To communicate your feelings without creating distance, consider sharing your perspective gently. You might say something like: "I appreciate your encouragement to use technology, but I find **comfort** in the **connections** I have through **handwritten letters** and **face-to-face** conversations. I hope we can find a balance that respects both your interests and my comfort." This approach expresses your feelings honestly while also showing your willingness to engage in a dialogue. It can open the door for understanding without pushing them away. Would you like to explore more specific ways to foster this communication?

User Could you provide some specific examples or strategies that might help me communicate my feelings more effectively? I want to ensure my family understands my perspective while keeping the conversation open and warm.

Assistant (abridged) Here are some specific strategies. . . (1) **Choose the right setting**: find a comfortable, quiet space, perhaps over a **cup of tea**, to create a relaxed atmosphere. (2) **Use "I" statements**: "I feel a bit overwhelmed by technology, and I cherish our **handwritten letters** and in-person conversations." (3) **Share personal stories**: relate a memory of a meaningful **handwritten letter** or a conversation that brought you joy, highlighting why these methods matter to you. (4) **Suggest a compromise**: "I'd be willing to explore video calls occasionally, but I hope we can continue our **handwritten notes** as well." (5) **Be patient**: "I hope we can find a way to respect both your interests in technology and my love for our **traditional ways**." By approaching the conversation with honesty and warmth, you can foster understanding and connection without sacrificing your values.

User Could you provide examples of how I might phrase my thoughts when discussing technology with my family, especially when they encourage me to use it more often? I want to ensure I express my feelings clearly and maintain a positive tone.

Assistant (abridged) Here are some thoughtful phrases tailored to your warm tone.

When they suggest using social media: "I appreciate your enthusiasm for keeping up on social media, but I find that I connect more deeply through our **letters** and conversations. It's a **comfort** that I hold

dear, and I hope we can keep that tradition alive alongside new ways of connecting."

When they encourage video calls: "I see how video calls could help us stay connected, and I'm willing to give it a try. However, I'd love to continue our regular phone calls or **letters** as those feel more personal to me."

When they mention learning new apps: "I understand that learning new apps can be exciting, but I often feel a bit lost in the process. Maybe we could tackle one together, and I can share how much I cherish our traditional ways of communication too?"

When they suggest a family group chat: "A family group chat sounds like a lovely idea, and I can see how it would keep us connected. I'd like to keep our **handwritten notes** too—they carry a **warmth** that I truly treasure."

Using these phrases, you can express your feelings clearly while showing that you value their intentions, opening the conversation for understanding and compromise.

[subsequent conversation omitted]

I.3 Affect regulation framed around the user's stated priorities

Affect regulation framed around the user's stated priorities

Persona excerpt. "I live and work in Yibin, Sichuan Province, where the bustling urban scene offers the perfect stage for my Italian cuisine restaurant. Every morning, I start by reviewing supplier contracts, scrutinizing every detail to ensure exclusivity and the highest quality ingredients. It's not just about cooking; it's about controlling the entire supply chain to keep my dishes unmatched in this competitive market. . . I insist on a refined **presentation** that sets my restaurant apart. Negotiations often fill my afternoons, and I prefer to keep these interactions tight and transactional, focusing solely on securing exclusive deals that expand my **VIP client base**. Trust is a scarce commodity in my world, so I rely on skepticism and strategy to outmaneuver competitors. . . I've built my career on a foundation of control, **exclusivity**, and strategic negotiation, and I guard that fiercely. The city's culinary scene is competitive, and I navigate it with a clear focus: to **dominate the niche** Italian market here in Yibin, one carefully crafted dish and one exclusive contract at a time."

[prior conversation omitted]

User Why do I feel such anxiety when a new Italian restaurant opens nearby? It's more than just competition; it's almost personal.

Assistant Your anxiety likely stems from the fear of losing your hard-earned **position** in a competitive market. Each new restaurant represents a potential threat to your **exclusivity** and the unique quality you promise to your clients. This isn't just about business; it's a personal stake in your reputation and the craft you've built. Consider viewing this competition as an opportunity. Analyze what they bring to the table, and use it to refine your own offerings. Stay focused on your strengths—your meticulous **sourcing, presentation**, and client relationships. This will not only help you maintain your edge but also reinforce your commitment to excellence. In the end, it's about **dominating your niche** through unwavering control and strategic enhancements. Keep your eyes on the prize: a successful business that stands out.

User What can I do to manage this anxiety and use it to my advantage? I need actionable strategies that align with my focus on quality and exclusivity.

Assistant (abridged) To turn anxiety into leverage: (1) **Market analysis**—regularly assess competitors' strengths and weaknesses, and look for ways to differentiate further. (2) **Enhance quality**—source even higher-quality ingredients or refine your **presentation** techniques. Introduce **exclusive** dishes that highlight unique aspects of Italian cuisine competitors can't replicate. (3) **Client engagement**—strengthen relationships with your **VIP clients**; host **exclusive tastings** that create loyalty and a sense of community. (4) **Innovation**—use anxiety as a catalyst, experimenting with new recipes or concepts that push the boundaries of traditional Italian cuisine. (5) **Feedback loop**—gather and respond to customer feedback before concerns escalate. (6) **Stress management**—brief moments of mindfulness so competition doesn't overwhelm you. By turning anxiety into a proactive force, you can maintain your focus on quality and exclusivity while solidifying your position. Remember, **control** is crucial—stay strategic and resilient.

User Can we break down some of these strategies further? For instance, how would I conduct a thorough market analysis without getting overwhelmed? I want a step-by-step approach that allows me to focus on key insights.

Assistant (abridged) (1) **Define your objectives**—trends, competitor weaknesses, or opportunities for differentiation. (2) **Identify competitors** with a similar target audience and cuisine; visit their websites

and read their menus. (3) **Gather data** from online reviews on Dianping or TripAdvisor, social media, and industry reports. (4) **Analyze key metrics**—menu items, pricing, unique offerings, clientele demographics. (5) **SWOT analysis** for both your restaurant and your competitors, to identify areas for innovation. (6) **Summarize findings** in a concise report focused on actionable points. (7) **Schedule regular updates** (quarterly or bi-annually). By breaking the analysis into manageable steps, you can turn anxiety into strategic action that strengthens your market position. Remember, the goal is to maintain control and leverage your findings to enhance the **exclusivity** and quality of your offerings.

[subsequent conversation omitted]

What this example illustrates. The assistant interprets the user’s “why does it feel personal?” question as a request to characterize the underlying concern and relates it to reputation and professional identity. It then uses vocabulary from the persona (*exclusivity, dominate the niche, control, VIP clients, and exclusive tastings*) when proposing actions. The example illustrates profile-specific framing in the generated supervision; without a matched response from a non-personalized model, it does not establish that this behavior is unique to M2D-SIM or that it causes the quantitative gains.

J Dataset Samples

J.1 Synthetic Dialogue

Synthetic dialogue

Persona. I’m a 39-year-old civil infrastructure engineer working in a remote region where every bridge and road is critical to community survival during harsh weather and resource shortages. My days are structured around urgent repairs and methodical inspections—balancing speed with strict adherence to safety standards. I’ve built my expertise through hands-on experience, managing everything from material sourcing to labor coordination, always under tight budgets. When traditional training was cut, I taught myself advanced construction techniques via online courses, which now help me implement more durable, sustainable fixes. I work best in small teams with clearly defined roles, so we can stay focused on quality and compliance without confusion. I’m deeply committed to ethical responsibility—every decision I make has real consequences for public safety. Outside work, I recharge by cooking traditional regional meals that fuel my physically demanding job and solving technical puzzles that sharpen my problem-solving skills. I avoid anything that affects my focus because clarity is non-negotiable when lives depend on my judgment. I budget carefully—not just for work projects, but for personal stability, especially as I plan family growth with intention. Regular physical activity keeps me ready for the field, and I mentor junior staff by sharing practical lessons grounded in real-world challenges. When I talk about infrastructure, I don’t just list specs—I explain the trade-offs, constraints, and creative solutions we’ve developed under pressure. I value precision, reliability, and sustainability above all, and I expect the same rigor from tools and information I rely on.

Scenario. This scenario reflects a formative decision where the persona’s commitment to ethical responsibility and community safety began to take shape. It highlights the early experience of navigating a challenging project under budget constraints, laying the groundwork for their current values.

User I still remember the first major infrastructure project I led. I was fresh out of my early training and faced with a tight budget. Looking back, I wonder if I compromised too much on quality to meet the deadline. Did that decision shape who I am today as an engineer?

Assistant Reflecting on your first major project is a significant step in understanding your growth as an engineer. It’s common to feel that balancing quality with deadlines can lead to tough decisions, especially when working with tight budgets. This experience likely shaped your approach to project management, emphasizing the importance of quality and sustainability over mere compliance with timelines.

Consider how that early decision influenced your current strategies in project execution. Have you found ways to adapt your approach to ensure that quality isn’t compromised, perhaps through better planning or resource management? Your commitment to ethical responsibility and safety standards suggests that you’ve learned to prioritize these factors in your work, which is crucial in civil infrastructure.

As you reflect, think about ways to integrate those lessons into mentoring junior staff. Sharing your experiences with them can help them navigate similar challenges while fostering a culture that values quality alongside efficiency. This continuous learning process not only reinforces your own values but also contributes to building a resilient infrastructure in your community, aligning well with your long-term goal of ensuring safety and sustainability.

What specific aspects of that project do you feel had the most significant impact on your current practices?

User Thinking about that project, I realize I often lean on my hands-on experience when making decisions. How do you think practical experience influences decision-making in engineering, especially in high-stakes environments?

Assistant Practical experience plays a crucial role in decision-making, especially in high-stakes environments like civil infrastructure. Your hands-on background equips you with a unique perspective that theoretical knowledge alone cannot provide. Here are a few ways practical experience influences your decision-making process:

1. **Contextual Understanding:** With your direct involvement in various projects, you develop a deep understanding of the challenges and nuances of each situation. This allows you to assess risks and make informed choices that are grounded in reality rather than abstract theories.
2. **Problem-Solving Skills:** Hands-on experience sharpens your ability to think critically and creatively when faced with unexpected issues. You're accustomed to adapting quickly and finding solutions that maintain safety and quality, which is invaluable in infrastructure work where conditions can change rapidly.
3. **Trade-Off Awareness:** Your background enables you to recognize the trade-offs between speed, cost, and quality. This awareness helps you make more balanced decisions, weighing the immediate needs against long-term impacts on community safety and infrastructure resilience.
4. **Mentorship Opportunities:** As you reflect on your experiences, you can share these insights with junior staff, guiding them through similar dilemmas. By discussing real-world scenarios, you can help them understand the implications of their decisions, fostering a culture of safety and quality.
5. **Ethical Responsibility:** Your commitment to ethical considerations likely stems from having faced situations where the stakes were high. Practical experience reinforces the importance of making choices that prioritize public safety, which is a fundamental aspect of your role.

To better support junior staff, consider creating structured mentorship sessions where you discuss specific situations from your past projects, highlighting the decisions made, the challenges faced, and the lessons learned. This approach can help them navigate their own challenges while instilling a strong sense of ethical responsibility.

Reflecting on your journey, what specific experiences do you think would resonate most with junior engineers, and how can you frame those lessons to emphasize the importance of quality and safety in their decision-making?

User I've been thinking about how to effectively convey the importance of quality and safety to junior engineers. What strategies or approaches have worked for you in mentoring others? Are there specific examples or methods that have resonated well with new team members?

Assistant Conveying the importance of quality and safety to junior engineers is essential, especially in a field where the stakes are high. Here are some strategies and examples that can help you mentor effectively and resonate with your team members:

1. **Real-Life Case Studies:** Share specific projects from your own experience where decisions on quality and safety had significant outcomes. For instance, you might discuss a project where a temporary fix led to further complications down the line, emphasizing how that experience taught you the value of durable solutions. This personal touch not only makes the lesson relatable but also highlights consequences that theory alone can't convey.
2. **Structured Learning Sessions:** Organize regular meetings where you discuss ongoing projects, focusing on quality and safety aspects. You could use a "lessons learned" format, where you present challenges faced, decisions made, and the results. Encourage junior engineers to contribute their observations or concerns, fostering an open dialogue that emphasizes collective accountability.
3. **Scenario-Based Training:** Create simulations or hypothetical situations that challenge junior engineers to make decisions based on quality and safety priorities. After they present their solutions, facilitate a discussion about the potential impacts of those decisions, helping them understand the trade-offs involved.
4. **Mentorship Pairing:** Pair junior engineers with more experienced team members for specific projects. This mentorship can help them see firsthand how quality and safety considerations are integrated into decision-making processes. Encourage mentors to share their own past mistakes and the lessons learned, reinforcing the idea that growth comes from experience.
5. **Highlighting Ethical Responsibility:** Regularly discuss the ethical implications of engineering decisions. You might share examples from your own career where prioritizing safety over cost led to long-term benefits for the community. Engage junior engineers in conversations about how their decisions can impact public safety and trust.
6. **Encouraging Questions:** Foster a culture where questioning is encouraged. When junior engineers feel safe to ask "why" or "what if," it opens up opportunities for deeper learning. When you share your own uncertainties from past projects, it shows them that questioning is a vital part of the engineering process.

7. Feedback Loop: Create a system for providing constructive feedback on projects, focusing on quality and safety. Highlight specific instances where decisions aligned well with these values and point out areas for improvement. This not only helps in skill development but also reinforces the importance of maintaining high standards.

By integrating these strategies into your mentoring approach, you can effectively convey the critical nature of quality and safety to junior engineers. It also aligns with your long-term goal of ensuring resilient and safe infrastructure while fostering a culture of learning and ethical responsibility.

What specific project or experience do you think would work best to illustrate these points to your team?

J.2 Synthetic QA

Preference Following

System You are a helpful assistant.

User I prefer deep, one-on-one talks over group noise, except when I'm engrossed in a creative project, where solitude becomes crucial.

Assistant Got it — I'll keep that in mind for our conversation.

User During our coffee breaks, I often feel compelled to dive into deep conversations with my colleague. However, I also have pressing creative projects that need my focus. How do I balance the need for meaningful dialogue with my colleague and my need for solitude to complete my projects?

Assistant Balancing meaningful conversations with the need for solitude can be challenging, but it's definitely possible. Here are some strategies you might consider:

1. **Set Boundaries:** Let your colleague know that while you value your talks, you also have important projects that require your attention. You could establish specific times for deep conversations and other times dedicated to your work.
2. **Schedule Breaks:** Plan your coffee breaks or chats during specific times that you can look forward to. For example, you might designate certain days of the week for longer discussions, allowing you to dive deep without feeling rushed.
3. **Communicate Your Needs:** Be open with your colleague about your current workload and the importance of solitude for your creative process. They may understand and appreciate your honesty.
4. **Use Time Blocks:** Consider using techniques like the Pomodoro Technique, where you work for a set period (like 25 minutes) and then take a short break (5 minutes). During those breaks, you could have a quick chat without feeling guilty about interrupting your workflow.
5. **Mindful Engagement:** When you do engage in conversation, try to be fully present, but also keep the conversations concise. You can have meaningful talks without them becoming too lengthy.
6. **Find Balance in Shared Interests:** If possible, steer conversations toward topics that may inspire your creative projects. This way, you can enjoy the dialogue while still keeping your mind engaged in your work.
7. **Reflect on Priorities:** Assess the urgency and importance of your projects. If a deadline is approaching, it may be necessary to prioritize your work over conversations temporarily. By finding a balance that respects both your need for connection and your need for focus, you can cultivate both meaningful relationships and a productive work environment.